Chapter 11: Data Analysis

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1 Introduction

This chapter covers factor selection methods and Twin-Gaussian Process models.

Citation: V. R. Joseph (2026). *Experimental Design for Data Science and Engineering.* CRC Press.

```
library(MASS)
library(first)
```

2 Section 11.1: Factor Selection and Ranking

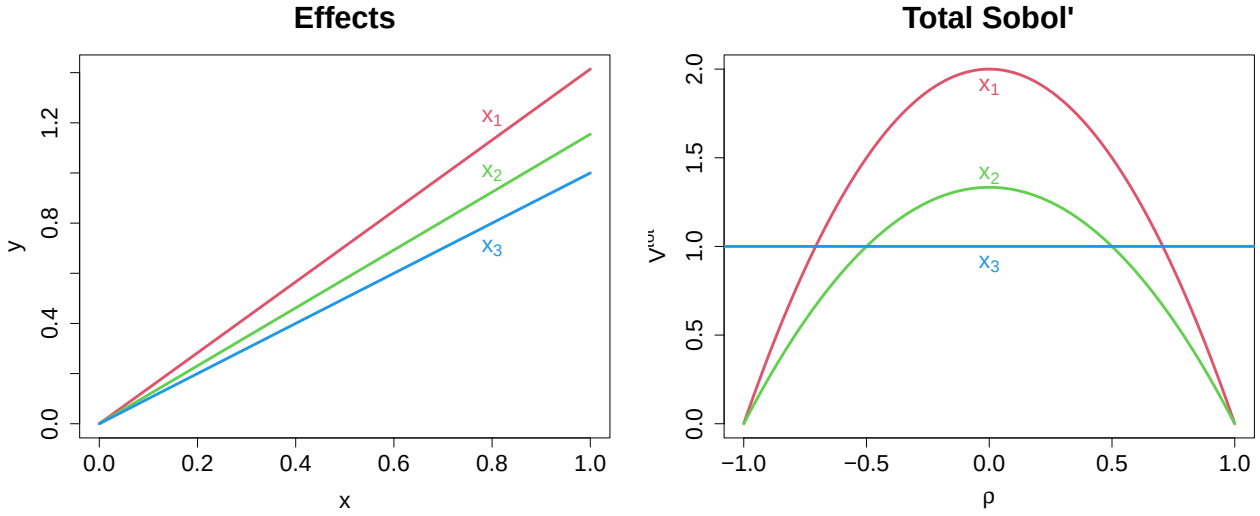
2.1 Effects and Total Sobol' Variance

```
par(mfrow = c(1, 2))

# Effects
curve(sqrt(2) * x, from = 0, to = 1, lwd = 3, main = "Effects",
      cex.main = 2, xlab = "x", ylab = "y", cex.lab = 1.5, cex.axis = 1.5, col = 2)
curve(2/sqrt(3) * x, from = 0, to = 1, add = TRUE, col = 3, lty = 1, lwd = 3)
curve(1 * x, from = 0, to = 1, add = TRUE, col = 4, lty = 1, lwd = 3)
text(0.8, 1.22, expression(x[1]), cex = 1.5, col = 2)
text(0.8, 1, expression(x[2]), cex = 1.5, col = 3)
text(0.8, 0.7, expression(x[3]), cex = 1.5, col = 4)

# Total Sobol' Variance
curve(2 * (1 - x^2), from = -1, to = 1, lwd = 3, main = "Total Sobol'",
      cex.main = 2, xlab = expression(rho), ylab = expression(V^tot),
      cex.lab = 1.5, cex.axis = 1.5, col = 2)
curve(4/3 * (1 - x^2), from = -1, to = 1, add = TRUE, col = 3, lty = 1, lwd = 3)
```

```
abline(h = 1, col = 4, lty = 1, lwd = 3)
text(0, 1.9, expression(x[1]), cex = 1.5, col = 2)
text(0, 1.4, expression(x[2]), cex = 1.5, col = 3)
text(0, 0.9, expression(x[3]), cex = 1.5, col = 4)
```




```
par(mfrow = c(1, 1))
```

2.2 FIRST Algorithm

```
set.seed(123)
N <- 1000
p <- 2
rho <- 0.9
Sigma <- diag(p)
Sigma <- rho^abs(row(Sigma) - col(Sigma))
D <- mvrnorm(n = N, mu = rep(0, p), Sigma = Sigma)
D <- cbind(D, rnorm(N))
y <- 3 * D[, 1] + 2 * D[, 2] + sqrt(3) * D[, 3]

cat("FIRST scores (all variables):\n")
```

```
## FIRST scores (all variables):
```

```
print(first(X = D, y = y))
```

```
## [1] 0.06015956 0.02575967 0.11355069
```

```
cat("\nFIRST scores (without x2):\n")
```

```
##
## FIRST scores (without x2):
```

```
print(first(X = D[, -2], y = y))
```

```
## [1] 0.8862397 0.1092233
```

```
cat("\nFIRST rankings (noise = FALSE):\n")
```

```
##
## FIRST rankings (noise = FALSE):
```

```r
print(first.rank(X = D, y = y, noise = FALSE))
```

```
## $ranking
## [1] 1 3 2
##
## $explained.variance
## [1] 0.8634445 0.9693164 1.0000000
```

2.3 Factor Selection Comparison: RF vs GP

```r
library(SPlit)
library(randomForest)
library(rkriging)

concrete <- read.csv(file = "../datasets/Concrete_Data.csv", fileEncoding = "UTF-8-BOM")
concrete.scale <- data.frame(scale(concrete))

set.seed(123)
rmse.RF <- rmse.GP <- matrix(0, nrow = 30, ncol = 2)

for (i in 1:30) {
  ind <- SPlit(concrete, splitRatio = 0.1)
  train <- concrete.scale[-ind, ]
  test <- concrete.scale[ind, ]

  # All factors - Random Forest
  a <- randomForest(x = train[, 1:8], y = train[, 9])
  pred <- predict(a, test[, 1:8])
  rmse.RF[i, 1] <- sqrt(mean((test$y - pred)^2))

  # All factors - Gaussian Process
  a <- Fit.Kriging(X = train[, 1:8], y = train[, 9], interpolation = FALSE,
                   kernel.parameters = list(type = "Gaussian"))
  pred <- Predict.Kriging(a, test[, 1:8])$mean
  rmse.GP[i, 1] <- sqrt(mean((test$y - pred)^2))

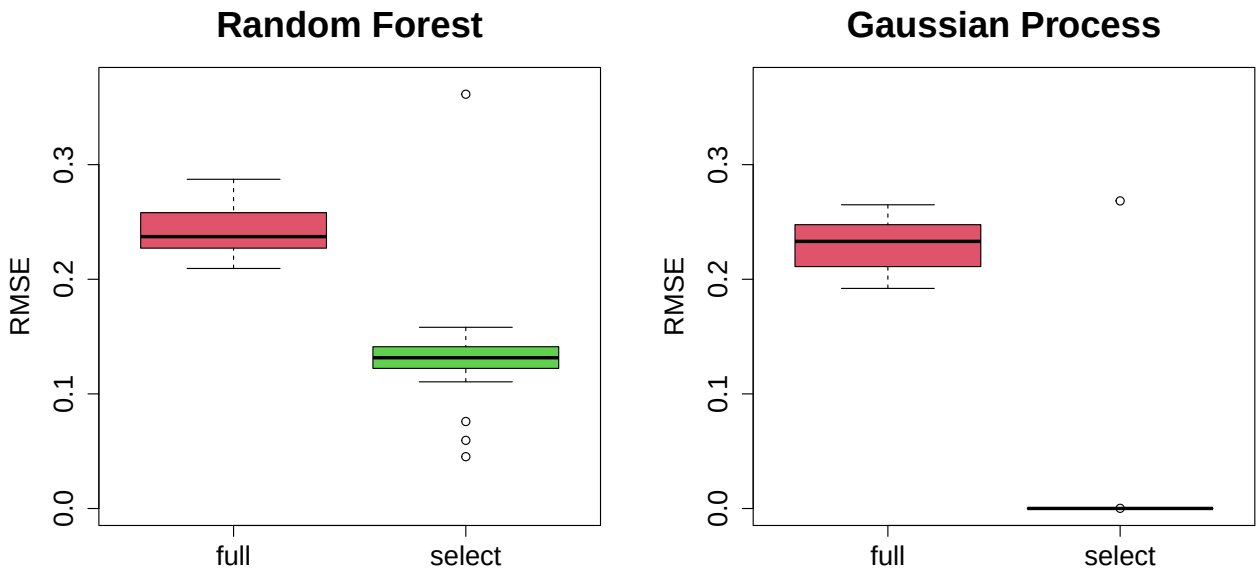
  # Selected factors
  sel <- first(X = train[, -9], y = train[, 9])
  sel <- (sel > 0)

  # Selected - Random Forest
  a <- randomForest(x = train[, sel], y = train[, 9])
  pred <- predict(a, test[, sel])
  rmse.RF[i, 2] <- sqrt(mean((test$y - pred)^2))

  # Selected - Gaussian Process
  a <- Fit.Kriging(X = train[, sel], y = train[, 9], interpolation = FALSE,
                   kernel.parameters = list(type = "Gaussian"))
  pred <- Predict.Kriging(a, test[, sel])$mean
  rmse.GP[i, 2] <- sqrt(mean((test$y - pred)^2))
}
```

```
## Optimizing <1 thread> [                    ] 1%Optimizing <1 thread> [                    ] 2%Optimi
## Optimizing <1 thread> [                    ] 1%Optimizing <1 thread> [                    ] 2%Optimi
```

```
## Optimizing <1 thread> [                    ] 1%Optimizing <1 thread> [                    ] 2%Optimi
```




```
par(mfrow = c(1, 1))
```

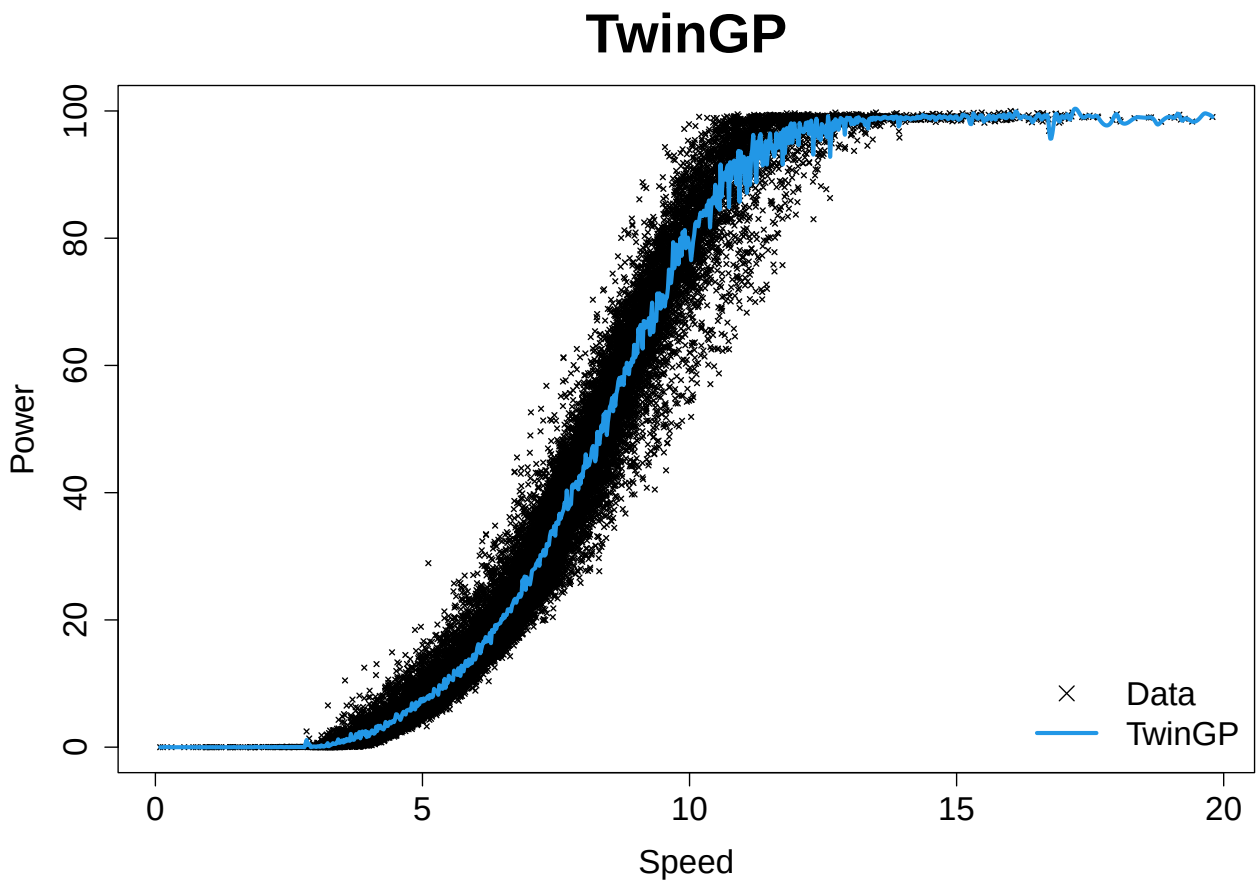
3 Section 11.2: Twin-Gaussian Process

```
library(twingp)

wind <- read.csv("../datasets/wind10.csv", header = TRUE)
D <- as.matrix(wind[, -1])
ev <- seq(min(D[, 1]), max(D[, 1]), length = 1000)

set.seed(123)
a.twin <- twingp(x = cbind(D[, 1]), y = D[, 2], x_test = cbind(ev))
pred.twin <- a.twin$mu

par(mar = c(5, 5, 4, 2))
plot(D, pch = 4, cex = 0.5, main = "TwinGP", xlab = "Speed", ylab = "Power",
     cex.main = 2.5, cex.lab = 1.5, cex.axis = 1.5)
lines(ev, pred.twin, col = 4, lwd = 3)
legend("bottomright", legend = c("Data", "TwinGP"),
       pch = c(4, NA), lty = c(NA, 1), col = c(1, 4),
       cex = 1.5, lwd = c(1, 3), bty = "n")
```



4 Summary

This chapter covered:

1. **Effects and Total Sobol' variance** - Theoretical framework for sensitivity
2. **FIRST algorithm** - Factor Importance Ranking and Selection using Total indices
3. **Factor selection comparison** - Random Forest vs Gaussian Process
4. **Twin-Gaussian Process (TwinGP)** - Robust GP for heterogeneous data